

UNICEF Analytics Challenge



Bui Minh Anh Pham, 1024739, anh.m.pham@aalto.fi
Eemeli Halminen, 907530, eemeli.halminen@aalto.fi
Manon Cartron Le Breton, 10215996, manon.cartron-lebreton@aalto.fi
Eevi Saari, 884158, eevi.saari@aalto.fi

Highlighted students are willing to participate in the competition.

UNICEF & the Sustainable Development Goals

Today's challenges for world's children are tightly linked to the 2030 UN's Sustainable Development Goals.

In this report, we will focus on the first 6 and 8th goals.

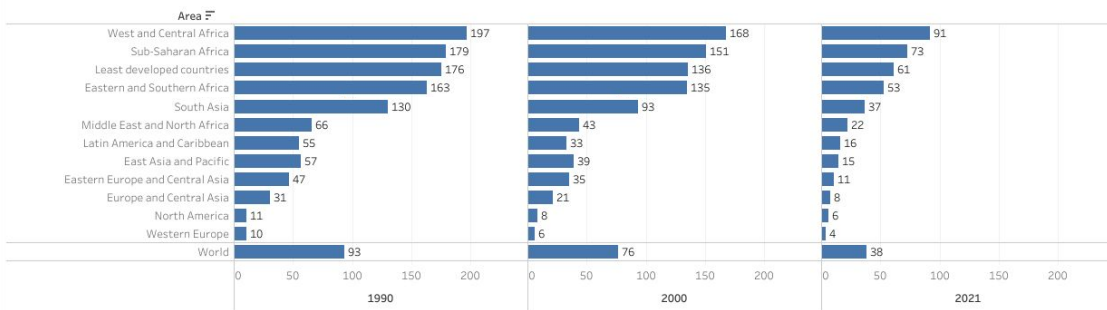


Child mortality, a global concern

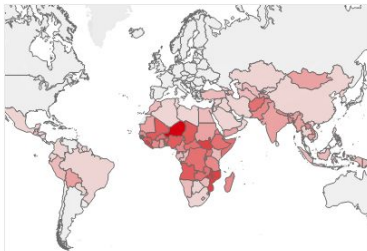
- Positive developments have been observed between 1990 and 2021: child mortality has decreased by 59% worldwide.
- Today, Africa is the continent most affected by child mortality.
- To understand child mortality rates, we are going to analyze different challenges faced by children today, regarding nutrition, health, education, child protection.



Evolution of under-five mortality rate by area - Per 1000

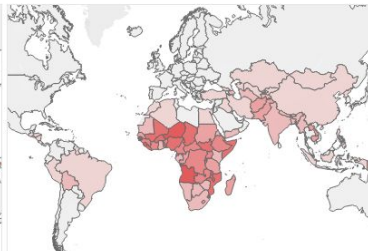


Under-five mortality rates in 1990



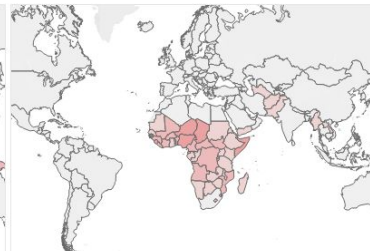
© 2024 Mapbox © OpenStreetMap

Under-five mortality rates in 2000



© 2024 Mapbox © OpenStreetMap

Under-five mortality rates in 2021



© 2024 Mapbox © OpenStreetMap

Under-five mortality rate - 1990 - Per 1000

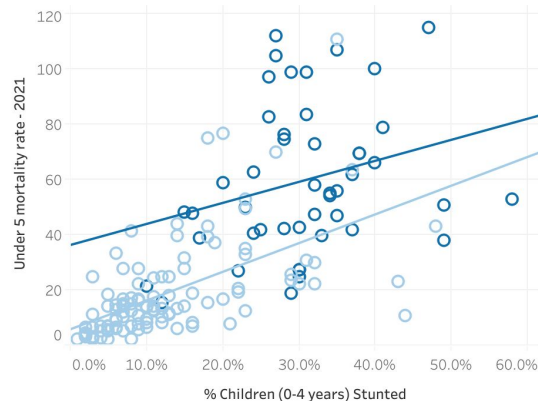


Malnutrition

- 22% of children aged 0-4 suffer from chronic malnutrition
- The most children affected in Africa
- Malnutrition is correlated with child mortality
- Women's malnutrition is correlated with low birth weights, a special concern in least developed countries



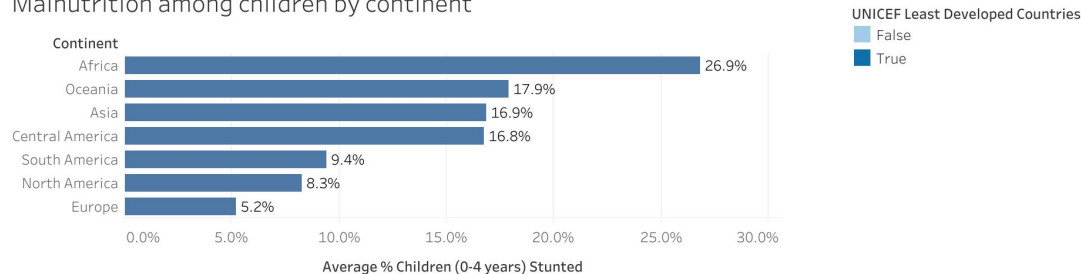
Malnutrition and child mortality



Women's malnutrition and low birthweight



Malnutrition among children by continent

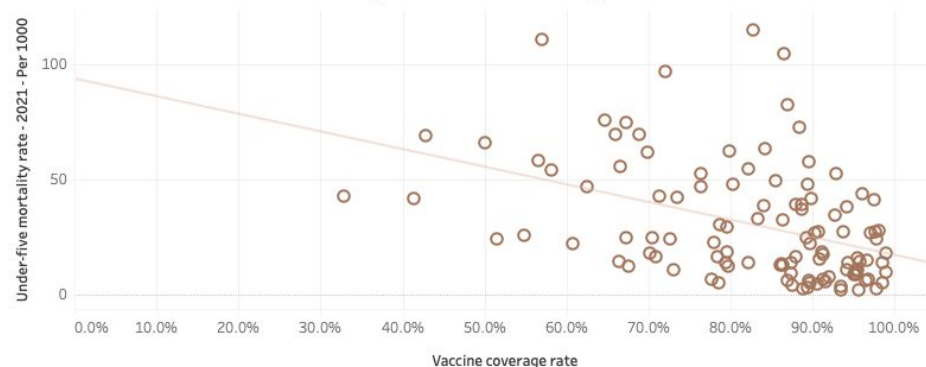


Health: The issues behind child mortality

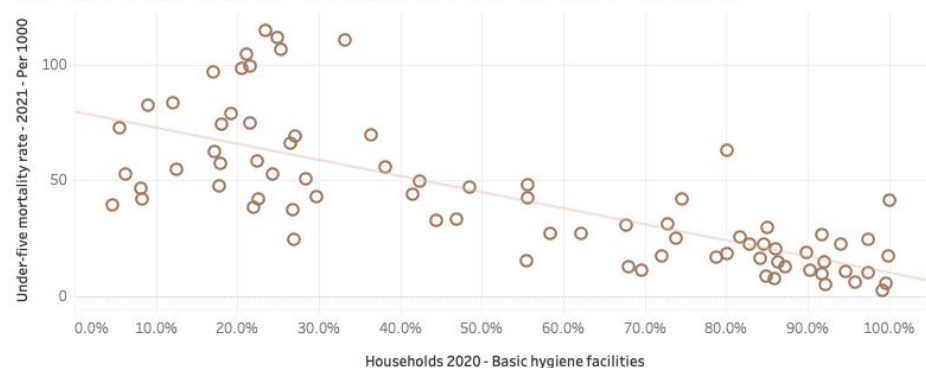
- Negative correlation between child mortality and vaccines coverage..
- Negative correlation between child mortality and presence of hygiene facilities in households.



Correlation between child mortality and vaccine coverage



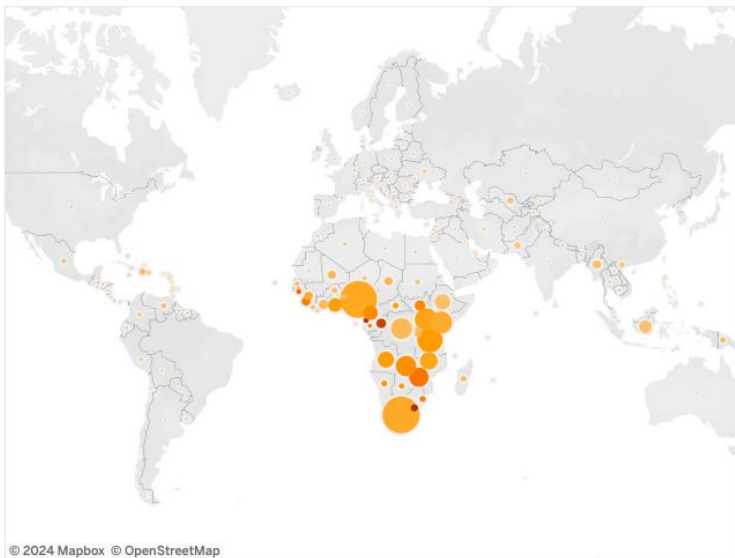
Correlation between child mortality and household hygiene facilities



Health: HIV/AIDS, a persisting challenge

- In Africa, AIDS related mortality is high for children.
- Most of the countries with the highest mortality rates related to AIDS don't have a high treatment rate for HIV in children.

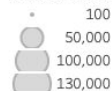
AIDS mortality and HIV prevalence in children



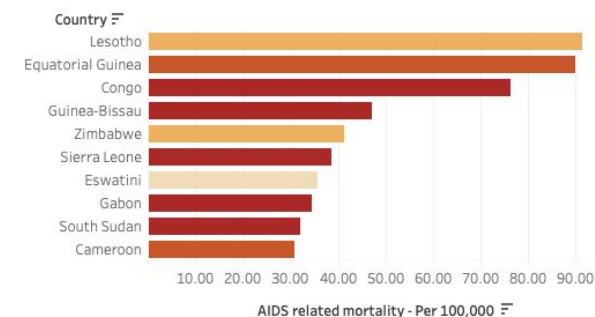
AIDS related mortality per 100,000 - Children 0-14

0.00 92.00

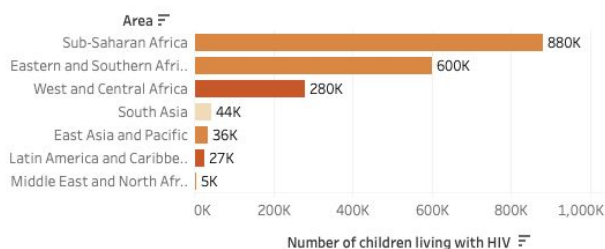
Number of children living with HIV - Children 0-14



AIDS related mortality (children 0-14)- By country



Number of children 0-9 living with HIV - By area



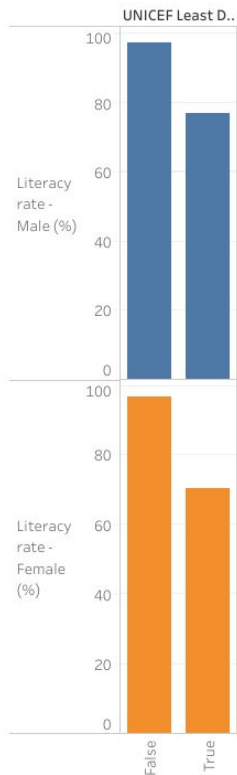
Children living with HIV receiving ART (%) - Children 0-14



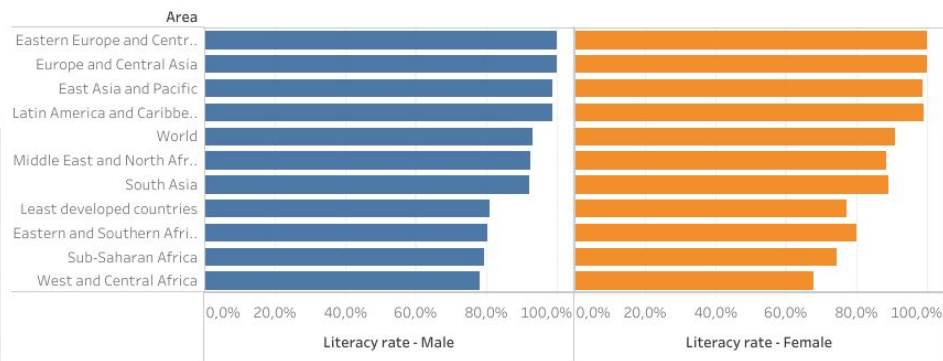
Education around the world

- Africa, especially West and Central Africa, has the lowest literacy rates and highest rates of children out of school at the primary level .
- In the least developed countries, literacy rates for females are substantially lower than for males.

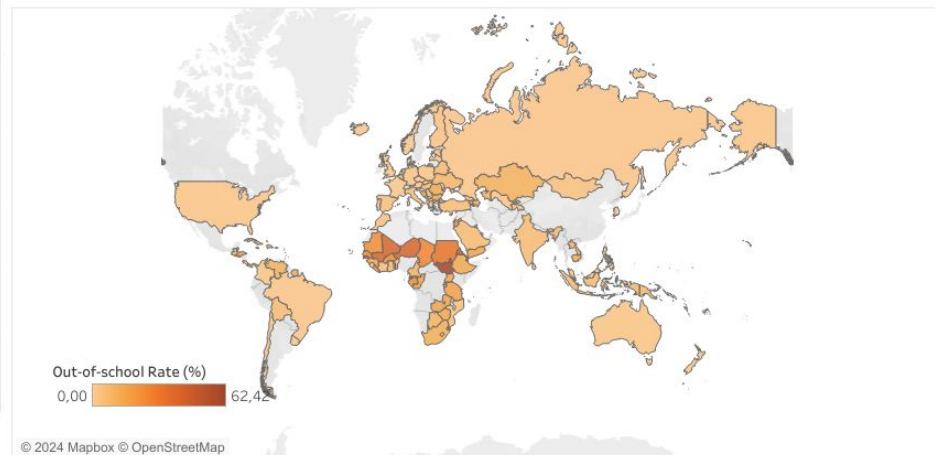
Literacy Rate in Least Developed Countries by Gender



Literacy Rate by Region and Gender



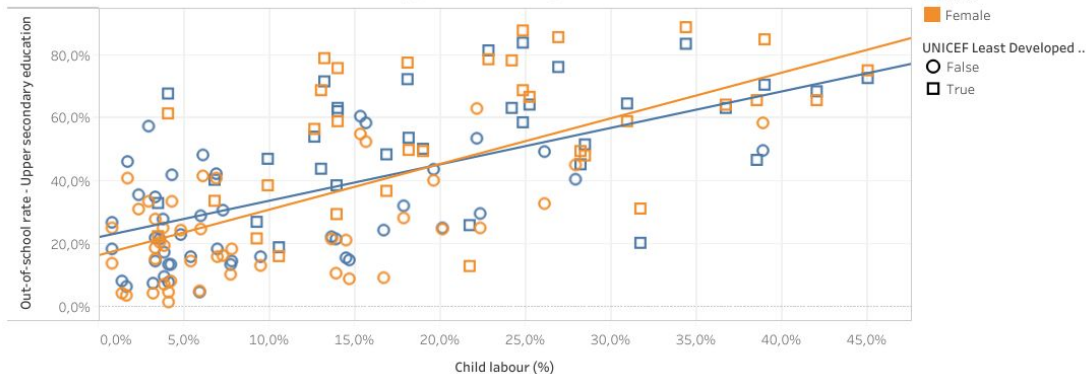
Out-of-school Rate



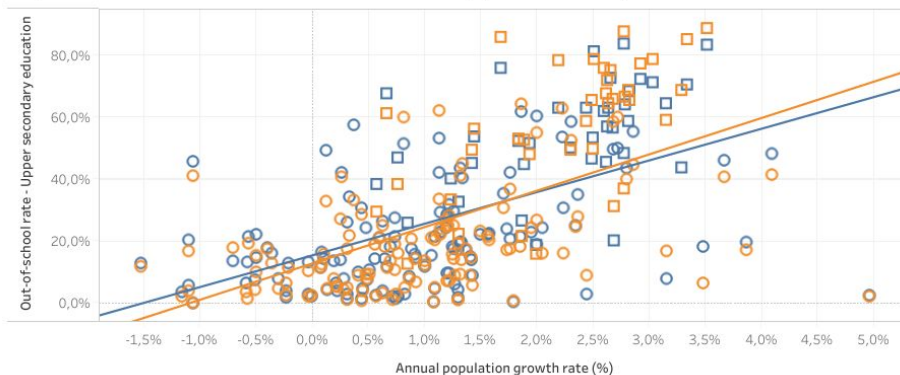
Education, Out-of-school rates

- Correlation between the child labour and the rate of children out of school in upper secondary education.
- Correlation between the population growth and the rate of children out of school in upper secondary education.
- In nations with low school completion rates, females typically have lower completion rates than males. However, in nations with high completion rates, the figures are nearly equal between genders.

Child Labour and Out-of-school rate Upper-secondary



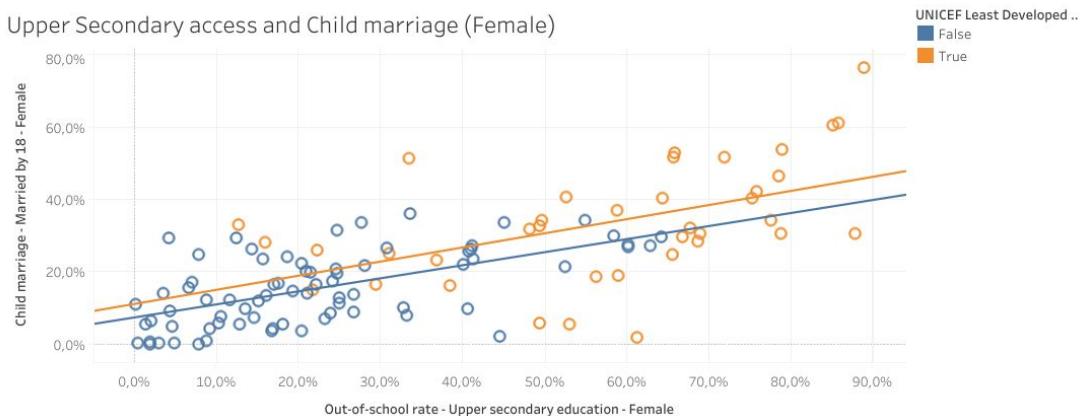
Population Growth and Out-of-school rate Upper-secondary



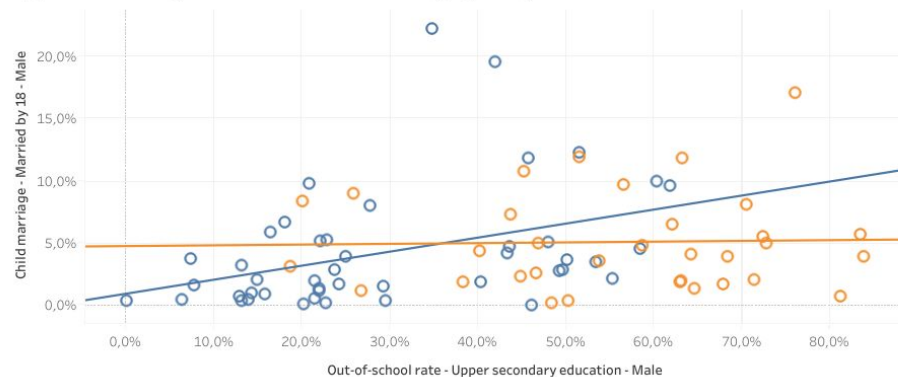
Education, the effect of Child marriage

- There is a strong positive correlation between child marriages among females and the lack of access to upper secondary education, particularly in the least developed countries.
- For males, a correlation between child marriages and lack of access to upper secondary education is observed primarily in developed countries.

Upper Secondary access and Child marriage (Female)



Upper Secondary access and Child marriage (Male)



Child Labour

- Overall there are more male child workers.
- Most prevalent in Africa and South Asia

Child Labour



Attitude toward Female Genital Mutilation

- In most countries with a high prevalence of female genital mutilation, over 50% of the population supports ending the practice.
- In nations like Burkina Faso and Eritrea, this figure rises to approximately 80%.



Attitude toward Female Genital Mutilation

